

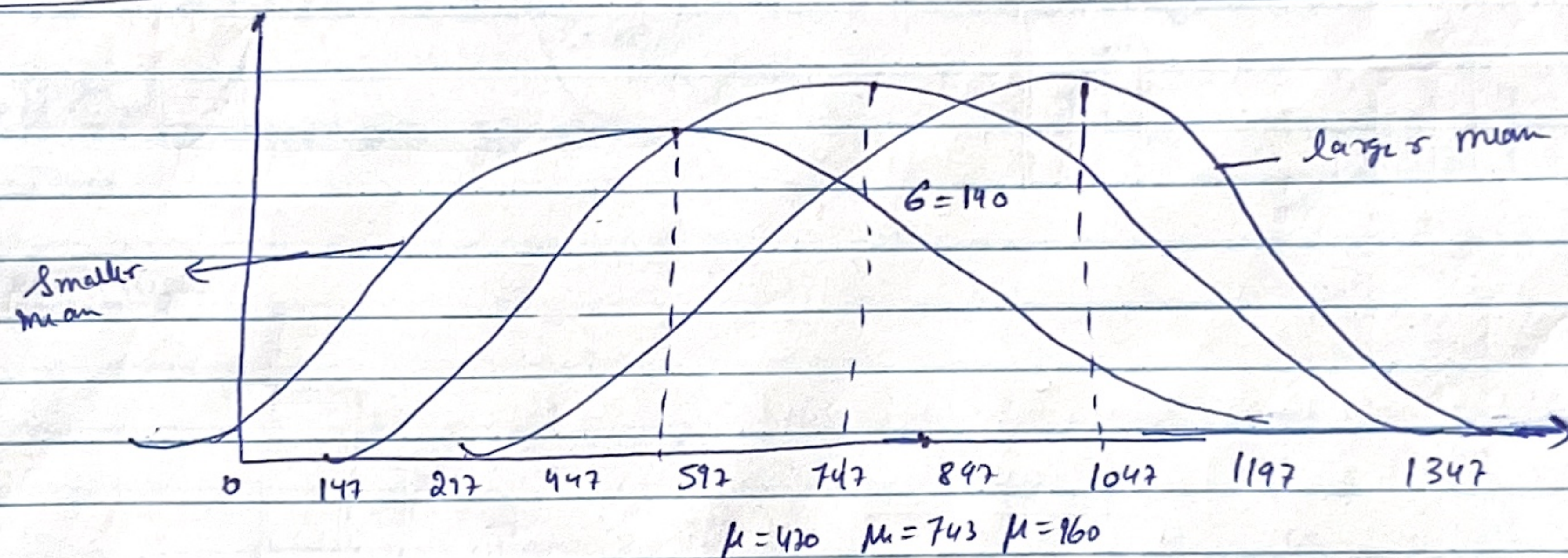
Normal Distribution

Reasons:

- Approximates wide range of random variables
- Distributions of sample means with large enough sample sizes could be approximate to normal
- All Computable statistics are elegant
- Heavy use of regression analysis

$N(\mu, \sigma^2)$
Normal Mean Variance
study for distribution

Control the S.D



Keep mean constant

Methods of Normalizing data

① Simple Feature Scaling: divides each value by max value for that feature

$$X_{\text{new}} = \frac{X_{\text{old}}}{X_{\text{max}}}$$

② Min-Max: Each value X_{old} subtract it from the minimum value of that feature then divides by the range of that feature.

$$X_{\text{new}} = \frac{X_{\text{old}} - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$$

③ Z-Score: Known as standard score. Each value subtract μ which is average of feature and then divide by SD (σ)

$$X_{\text{new}} = \frac{X_{\text{old}} - \mu}{\sigma}$$

Categorical Variables

Most statistical models cannot take categorical (objects / strings) as F/E

➡

Car	Fuel	...
A	Gas	...
B	Diesel	...
C	Gas	...
D	Gas	...

Add dummy variable for each unique category

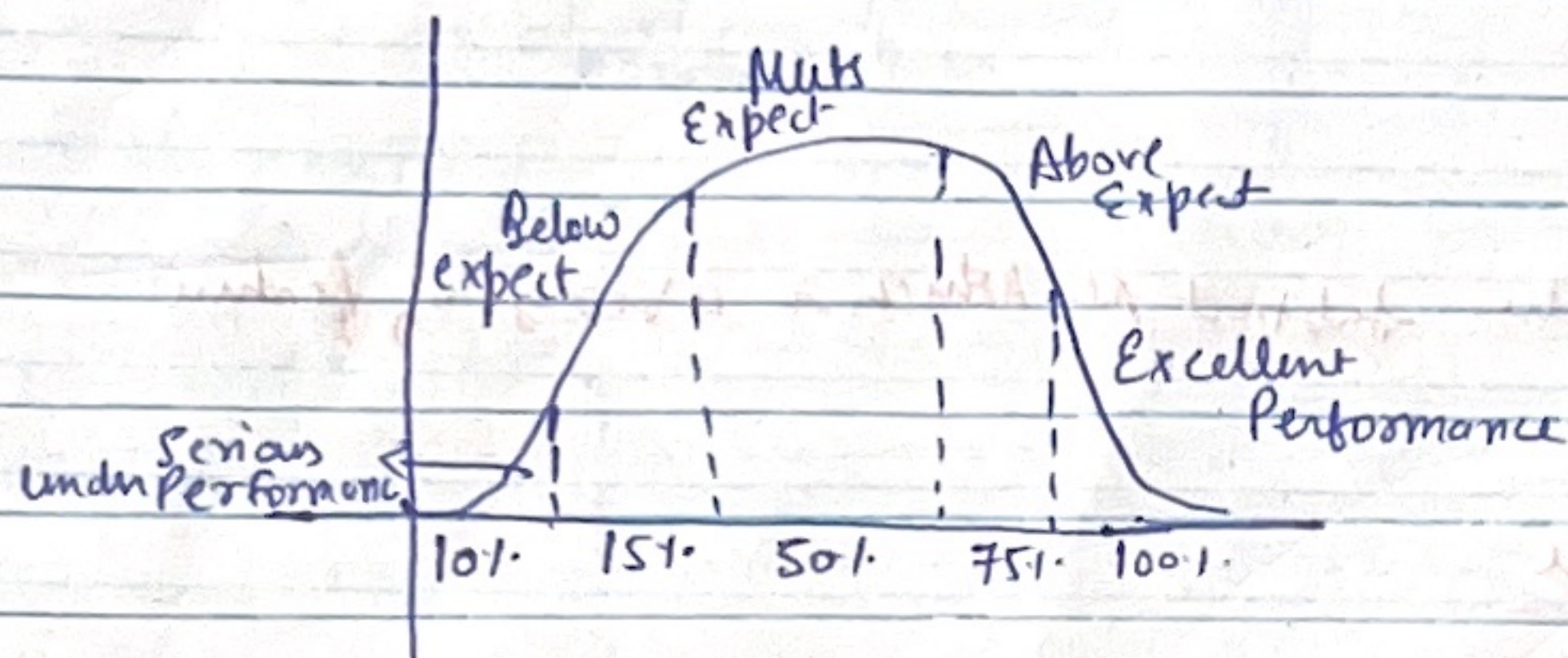
Car	Fuel	gas	diesel
A	G	1	0
B	D	0	1
C	G	1	0
D	G	1	0

One hot encoding

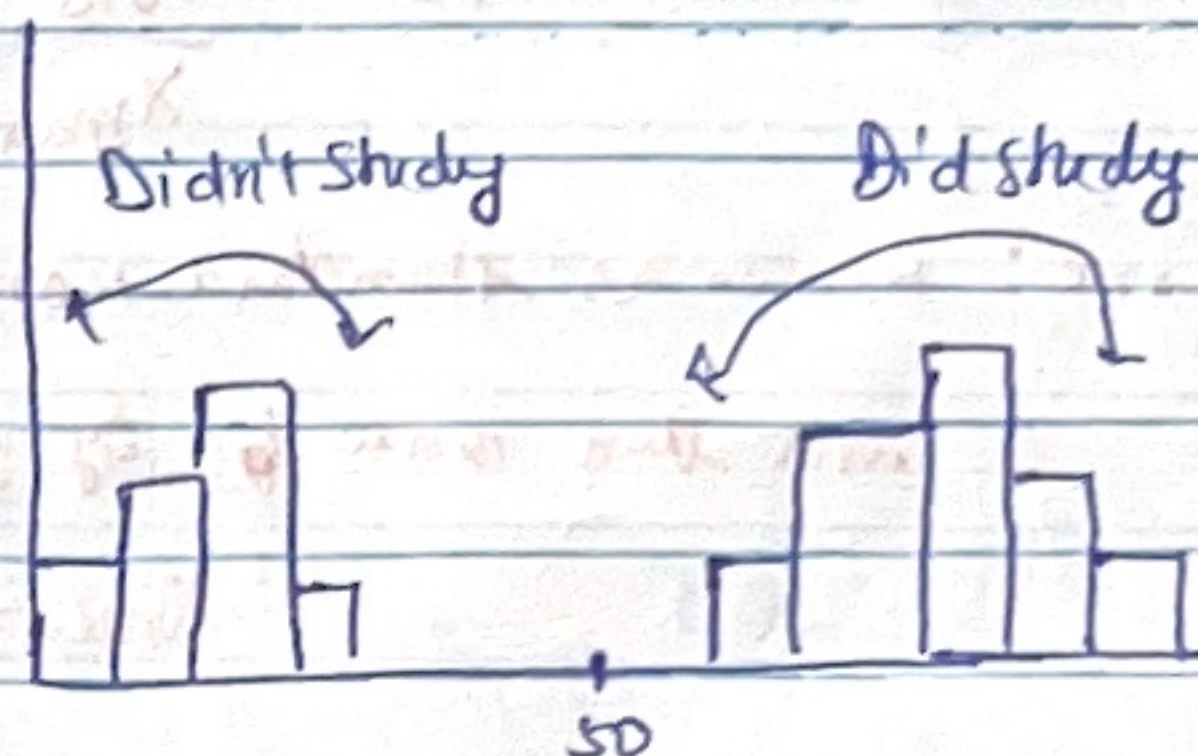
Histograms

- Used for Quantitative Variable
- ~~Bell~~ Bell shaped Curve (IQ, SAT)
- Skewed right (Annual income)
- Skewed left (score on easy exam)
- Bimodal (Polarized opinion)

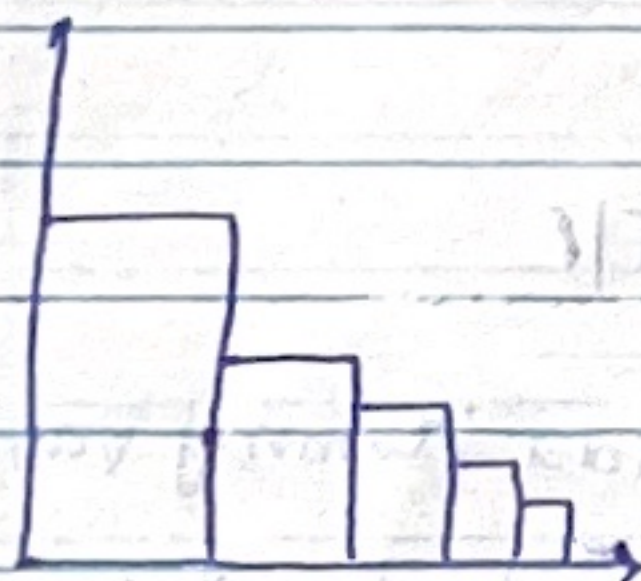
1) Bell-shaped



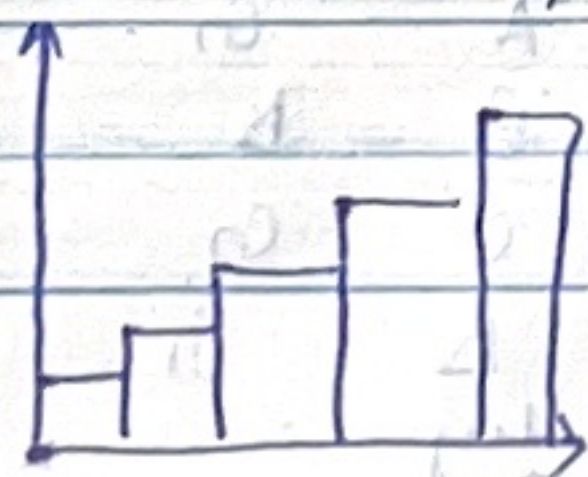
2) Bimodal



3) Skewed Right



4) Skewed Left



Correlation doesn't imply causation.

P-Values

$< 0.01 \rightarrow$ Strong certainty

$< 0.05 \rightarrow$ Moderate certainty

$< 0.1 \rightarrow$ Weak certainty

$> 0.1 \rightarrow$ No certainty

Strong Correlation

Correlation coefficient close to 1 or -1

P-value less than 0.001

